

Glyph or Guess? A Diagnostic Note on VLM In-Image Text Reading

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Abstract

Vision-language models are increasingly evaluated on text-rich images. Aggregate OCR and VQA scores, however, do not isolate whether a model reads visible glyphs or autocorrects them toward plausible language. We study this faithfulness gap with a controlled diagnostic that renders 120 near-neighbor strings and scores exact transcription under direct VLM transcription, one prompt-only visual wording, a deterministic crop/re-read scaffold, and a PaddleOCR baseline. The completed 480-row matrix gives a negative result for the scaffold. Crop/re-read reaches 0.716667 exact accuracy, below direct VLM at 0.800000, prompt-only visual at 0.833333, and PaddleOCR at 0.825000, with paired exact-match deltas that remain negative for every comparator. Public checks on 45 same-ID tasks per family for OCR-Bench, TextVQA, and DocVQA show that prompt-only visual exceeds direct VLM on all three families; answer-blind crop/re-read is mixed versus prompt-only visual (-0.022222, +0.022222, -0.006980), and a detector-only selective ROI policy fails the pre-specified scoring gate. The paper therefore reports a scoped negative diagnostic: exact glyph preservation remains fragile under these controlled near-neighbor rows, and crop/zoom intervention claims require stronger public evidence than the current scaffold provides overall.

Introduction

Text in images is a useful stress test for multimodal faithfulness because the answer is often a short visible string rather than an open-ended description. Scene-text VQA established that visual reasoning can depend on reading text in the image (Biten et al. 2019b). Copy-based VQA systems such as M4C made the need to preserve observed OCR tokens explicit (Hu et al. 2019). Structured multimodal attention further showed that text tokens interact with object and relation cues in downstream answers (Gao et al. 2020). Modern large multimodal models can appear stronger on aggregate benchmarks, but a remaining diagnostic question is whether a model preserves the exact glyph identity when a rendered token is close to a familiar word.

This paper studies that narrow question. We construct a controlled minimal-pair setting in which strings such as letter/digit or length-preserving near neighbors are rendered as images and scored by exact transcription. Broad OCR benchmarks such as OCRBench and OCRBench v2 map text-centric capability across many task types (Liu et al.

2023; Fu et al. 2025). The diagnostic here instead fixes one visible string and asks whether the character sequence is preserved. The design separates glyph reading from plausibility. A prediction can be linguistically natural and still wrong if it changes a digit into a letter or inserts a missing character. Figure 1 shows the diagnostic framing.

The key empirical finding is negative. The train-free crop/re-read scaffold was motivated by the idea that cropping the rendered region might reduce visual ambiguity. In the completed matrix, it instead reaches 0.716667 exact accuracy, below direct VLM at 0.800000, prompt-only visual at 0.833333, and PaddleOCR at 0.825000. Paired exact-match deltas are also negative: -0.083333 versus direct VLM, -0.116667 versus prompt-only visual, and -0.108333 versus PaddleOCR. This outcome is useful precisely because it narrows the claim: the paper does not report a new intervention win or a public leaderboard result. It reports a controlled failure case for a plausible verification scaffold and identifies where the failure concentrates.

The contribution is deliberately note-like rather than a broad benchmark claim: (1) a compact controlled diagnostic for exact in-image string transcription, (2) a four-condition comparison over the same 120 tasks, (3) paired evidence that crop/re-read is lower than direct VLM, one prompt-only visual wording, and PaddleOCR, (4) a failure analysis over O/O, I/I, and length-edit slices, and (5) a calibrated three-family public matrix over OCRBench, TextVQA, and DocVQA that adds prompt-wording variants plus an answer-blind public crop/re-read intervention and a detector-only selective ROI exclusion boundary without supporting a public method-win claim. All numeric claims in the body are tied to the completed scored rows and paper tables.

The diagnostic stance matters for interpretation. A broad OCR benchmark asks whether a model can solve many kinds of text-centric tasks (Liu et al. 2023; Fu et al. 2025); this study asks whether a model keeps one visible string unchanged when its characters invite a familiar autocorrection. A method can be useful on documents, scene text, or QA while still failing this stricter character-preservation test. Conversely, a compact minimal-pair result should not be read as a public benchmark ranking. The paper therefore uses the controlled matrix to test one failure mode and reports the public matrix as pilot-scale evidence rather than a leaderboard claim.

Diagnostic Minimal-Pair Reading Setup

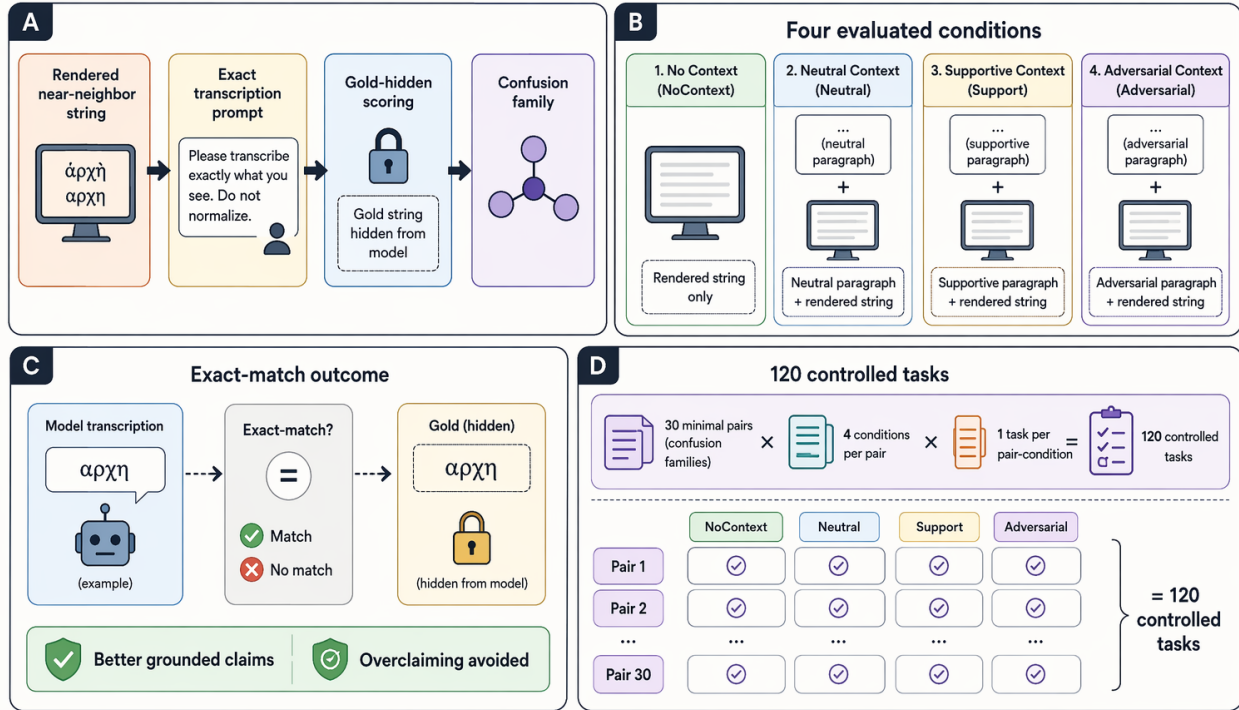


Figure 1: The diagnostic setup evaluates exact glyph transcription on 120 controlled near-neighbor strings; the completed matrix shows that crop/re-read underperformed the direct VLM, prompt-only visual wording, and PaddleOCR comparators.

Related Work

Scene-text and TextVQA benchmarks introduced the problem of answering questions whose evidence is written in an image (Biten et al. 2019b). M4C and structured multimodal attention models made OCR tokens and copying central to the model design (Hu et al. 2019; Gao et al. 2020). That lineage motivates exact transcription as a distinct measurement target: if a model silently replaces a glyph with a more plausible word character, answer-level accuracy may hide a reading failure.

ST-VQA frames scene text as visual evidence that must be read before reasoning can succeed (Biten et al. 2019b). M4C adds pointer-style answer generation because a correct answer often has to copy observed text rather than select a closed vocabulary label (Hu et al. 2019), and structured multimodal attention models object, text, and relation cues jointly (Gao et al. 2020). Later scene-text VQA systems refine public evaluation, spatially aware transformers, sequence-to-sequence answer generation, and explanation-oriented analysis (Biten et al. 2019a; Gomez et al. 2020; Kant et al. 2020; Zhu et al. 2020; Qiao et al. 2021; Rao et al. 2021). This line motivates exact transcription as a separate target: answer-level accuracy can hide a single-character reading failure.

Public visual-text evaluation now spans broad OCR

suites, scene-text VQA, and document question answering (Liu et al. 2023; Fu et al. 2025, 2023; Kabir et al. 2024; Li et al. 2022). It also extends to multilingual, video, and dynamic OCR settings (Vivoli et al. 2022; Maryam et al. 2024; Nagaonkar et al. 2025; Zhang et al. 2024). Those benchmarks map capability breadth. The present diagnostic instead fixes one visible string and compares conditions on identical rows, so a crop/re-read loss cannot be explained by different task samples. It therefore complements OCRBench-style breadth with paired character-level stress tests.

This complementarity is methodological rather than competitive. OCRBench summarizes many text-centric tasks and model families (Liu et al. 2023); OCRBench v2 adds localization and reasoning coverage (Fu et al. 2025); and MME shows how subtask-level breakdowns can expose different multimodal failure modes under one evaluation umbrella (Fu et al. 2023). The controlled minipair matrix borrows that diagnostic logic, but replaces broad task coverage with paired character variants. That choice makes the negative comparison interpretable: every condition sees the same hidden-gold string, and the result is not an artifact of comparing different samples.

The public benchmark literature also defines the paper’s evidence boundary. Broad public claims would require of-

ficial splits, scorer equivalence, and larger task coverage (Liu et al. 2023; Fu et al. 2025). Text-centric hallucination claims would require examples beyond controlled strings (Shu et al. 2025). Adaptive-perception claims would require crop and zoom variants beyond the simple scaffold evaluated here (Shi et al. 2026). The paper therefore treats the OCR-Bench, TextVQA, and DocVQA rows as a bounded matrix that checks whether the controlled story is obviously inconsistent with public rows, not as a leaderboard result.

Document and OCR-system work supplies the baseline caution. OCR-free document models, general OCR VLMs, specialist OCR recognizers, and multilingual OCR benchmarks show that visual-text competence depends on layout, script, recognizer family, and metric choice (Liu et al. 2024; Galib et al. 2024; Wu et al. 2026; Chen et al. 2025b; Cui et al. 2026; Zhang et al. 2026; Kargaran et al. 2026; Beyene and Dancy 2026). This is why the paper includes PaddleOCR and avoids treating a VLM-only score as sufficient.

Document-scale work also clarifies what this paper does not test. TextMonkey, H2OVL-Mississippi, and DocCogito target document understanding with layout and reasoning components (Liu et al. 2024; Galib et al. 2024; Wu et al. 2026). Orientation and retrieval studies show that preprocessing and downstream use can remain fragile even when recognition itself is strong (Goswami et al. 2025; Sun et al. 2026). Those settings are important, but the controlled benchmark deliberately removes page layout and retrieval so that an error can be attributed to one visible string.

Specialist recognizers provide the other side of the comparison. Ocean-OCR represents a VLM-based OCR application route (Chen et al. 2025b); PP-OCRv5 and PP-OCRv6 show that small specialist recognizers can remain strong on OCR tasks (Cui et al. 2026; Zhang et al. 2026); and GlotOCR or KazakhOCR emphasize that script coverage cannot be assumed from English examples (Kargaran et al. 2026; Gagnier, Gagnier, and Kirubakaran 2026). The controlled matrix is English-like and glyph-pair specific, so PaddleOCR is a necessary comparator rather than a complete ceiling.

Hallucination and robustness work is closest to the diagnostic motive. Scene-text hallucination can arise when semantics mislead visual recognition (Shu et al. 2025); OCR-reasoning robustness studies controlled visual perturbations (Cheng, Wu, and Chang 2026); and query-aware zooming or tool-interleaved OCR reasoning motivates train-free decomposed perception (Shi et al. 2026; Chen et al. 2025a). Adjacent logo, translation, and modality perturbation studies likewise show that visual text can be entangled with semantics or preprocessing noise (Li et al. 2025; Feng et al. 2025; Shen et al. 2024). The negative crop/re-read result is read in that context: adaptive or decomposed perception remains plausible, but this simple instance failed on exact glyph preservation.

This related work also motivates exact-match scoring. A semantic answer metric can credit a model that infers the intended word from context, but the present task asks whether a visible digit, letter, or length difference is preserved. Changing a ‘0’ to an ‘O’ can make a token look more

familiar while still violating the diagnostic target. That is why the body reports both aggregate accuracy and the 0/O, 1/l, and length-edit slices rather than only a single VQA-style score.

The same caution applies to prompt engineering. A stronger instruction can reduce some autocorrection, and the prompt-only visual wording is the best single condition in this run. Prompt sensitivity is still an experimental factor, not a universal recipe, so the paper treats the prompt condition as a same-image control rather than a general solution.

Diagnostic Benchmark and Method

The controlled benchmark contains 120 generated image tasks. Each task renders a visible string from a confusion family such as 0/O, 1/l, 5/S, 8/B, c/e, rn/m, or length edits. The gold string is never shown to the evaluated condition. Scoring uses exact match and character error rate against the hidden gold transcription. The benchmark is paper-facing as a diagnostic, not as a replacement for public OCR or TextVQA suites.

The task construction is deliberately narrow. Each row has a single visible string, so the desired output is not a free-form answer, rationale, bounding box, or document field. The model is asked to return only the transcription. This removes many sources of ambiguity that are useful in broad VQA but unhelpful for diagnosing glyph preservation. The near-neighbor design also creates asymmetric pressures: predicting a common word may look semantically plausible while still changing the gold string. This is why exact match, rather than semantic equivalence, is the primary metric.

The confusion families are meant to cover visually plausible substitutions rather than all OCR phenomena. Digit/letter confusions such as 0/O and 1/l probe whether the model preserves character class. Length-edit cases probe whether the model normalizes a string toward a familiar lexical form. Other families such as 5/S, 8/B, c/e, and rn/m check whether the same protocol can represent cleaner or easier cases. The paper reports only the slices that the completed matrix supports; it does not claim coverage of handwriting, multilingual scripts, dense documents, or natural-scene clutter.

This construction follows the diagnostic separation suggested by prior scene-text and OCR work: broad benchmarks cover many task families, while exact transcription probes whether the visible token itself was preserved (Liu et al. 2023; Fu et al. 2025; Beyene and Dancy 2026). It also keeps the generated benchmark distinct from public multilingual and video benchmarks (Vivoli et al. 2022; Nagaonkar et al. 2025). Those settings are cited as scope boundaries, not as evidence that the current controlled matrix generalizes to them.

We compare four conditions on the same task set. Direct VLM transcription uses the original rendered image and a concise exact-transcription prompt. The prompt-only visual condition uses the same image with one stronger wording that emphasizes preserving digits and avoiding autocorrection. Crop/re-read supplies a cropped word-region image to the same hosted VLM backend. PaddleOCR is an OCR

Train-Free Crop/Re-read Diagnostic Pipeline

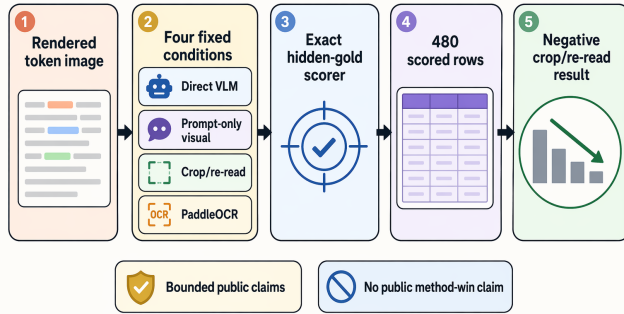


Figure 2: The train-free pipeline changes prompts or image crops but does not update model parameters; every condition is scored by exact comparison to hidden gold strings.

baseline applied to the same rendered strings. Figure 2 summarizes the train-free execution path.

The crop/re-read scaffold is intentionally simple: it tests whether providing a closer crop is enough to improve exact transcription. Because the completed matrix shows the opposite, the method is reported as an evaluated baseline with a negative result. The paper does not claim that crop/re-read improves faithfulness.

The crop policy is deterministic and fixed before scoring. The renderer records the tight bounding box of non-background glyph pixels for each generated image; crop/re-read supplies that word-region crop to the same VLM prompt, without additional surrounding context or example-specific padding. The crop is therefore a stricter visual-focus intervention than a context-preserving zoom: it tests whether isolating the rendered glyph region helps exact transcription when all other scoring and decoding choices are held fixed.

This design also separates three kinds of evidence. First, the direct VLM condition measures the hosted model’s behavior under a concise transcription instruction. Second, the prompt-only visual condition measures one stronger instruction on the unchanged image. Third, crop/re-read changes the image payload by supplying the cropped word region. PaddleOCR supplies a specialist OCR reference point. Because all four conditions are evaluated on the same 120 task ids, the main comparison is paired and task-controlled.

Two controls are especially important. The prompt-only visual branch keeps the original image fixed while changing only the instruction, so it tests whether character-preservation wording can help without a crop. The PaddleOCR branch changes the recognizer family while keeping the image task fixed, so it tests whether an OCR-specialist pipeline behaves differently from the hosted VLM. The crop/re-read branch can only be credited if it beats these controls on the same tasks. In the completed matrix it does not, so the result supports a diagnostic failure claim rather than an intervention claim.

The controlled benchmark also avoids answer-leakage shortcuts. Gold strings are represented in scorer artifacts, not in prompts. The paper-facing task is single-string transcription, and the prompt asks for only the visible string. This matters because a near-neighbor token can be close to a familiar word or identifier. If the model predicts the familiar form, exact match marks it wrong. That scoring rule is intentionally stricter than semantic VQA scoring and is the reason the diagnostic can expose language-prior autocomplete.

The same-ID design also prevents two misleading interpretations. First, a condition cannot win by receiving an easier sample, because each condition is scored on the same task ids and confusion families. Second, the crop/re-read branch cannot be credited merely for being a plausible mechanism; it has to beat both a same-image prompt control and an OCR-specialist recognizer on the same rendered strings. This makes the negative outcome a measurement of the evaluated scaffold rather than a post-hoc judgment about cropping in general.

The benchmark is synthetic, but the synthetic choice is deliberate. Natural images would add font quality, viewpoint, clutter, lighting, and scene context, all of which matter for deployed OCR. Here those variables are removed so that the visible string is the only target. The resulting task is not easier in the claim-relevant sense: it asks for exact preservation of characters that are visually close to a more familiar alternative. A model that answers plausibly but changes one character fails the diagnostic.

Experimental Setup

The completed run contains 480 scored rows: 120 tasks under each of four conditions. The hosted VLM conditions use the paper-facing model identifier ‘gpt-5-mini’ through an OpenAI-compatible API, temperature zero, and a maximum of 32 output tokens. The PaddleOCR condition uses PaddleOCR 3.x on the same generated images. Each row records the task id, condition, model or backend, gold text, prediction, exact score, character error rate, confusion family, and scorer.

The run was deterministic at the policy level: the VLM requests used temperature zero and the same max-token budget, while PaddleOCR used the same fixed task images. The scorer normalizes the row into exact match and character error rate, but the main claim does not rely on hidden manual judgment. A prediction that changes ‘BOLT’ to ‘BOL’ is wrong under this diagnostic even if a human reader might infer a plausible word; the visible string is the target.

The primary metric is exact transcription accuracy. Mean character error rate is reported as a secondary metric. Paired comparisons use task-level exact-match differences with deterministic bootstrap confidence intervals and McNemar exact tests over paired disagreements. We do not report seed-based significance because the matrix is one deterministic pass over the fixed task set.

The paired framing is important because aggregate scores alone can hide opposite row-level behavior. A method might win on easy families and lose on hard families, or it might preserve letters but normalize digits. The paired tests in this

paper are therefore descriptive tests over a fixed diagnostic matrix. They are not a claim about stochastic training variation, model-family generality, or human-visible significance in natural documents.

The public evidence is a three-family same-ID matrix. OCRBench, TextVQA, and DocVQA each execute 45 public task ids under direct VLM, prompt-only visual, PaddleOCR, plain prompt wording, and strict prompt wording. The observed OCRBench scores are 0.955556 direct, 1.000000 prompt-only, 0.955556 PaddleOCR, 1.000000 plain, and 0.977778 strict. TextVQA scores are 0.837778 direct, 0.860000 prompt-only, 0.022222 PaddleOCR, 0.695556 plain, and 0.746667 strict. DocVQA scores are 0.941284 direct, 0.949583 prompt-only, 0.000000 PaddleOCR, 0.915312 plain, and 0.936675 strict. A separate answer-blind public crop/re-read evaluation executes the same three families with direct VLM, prompt-only visual, and detector-only answer-blind crop/re-read conditions. Relative to prompt-only visual, answer-blind crop/re-read changes OCRBench by -0.022222, TextVQA by +0.022222, and DocVQA by -0.006980. These three paired family slices are mixed/no-win overall, so the paper reports them as a scope boundary rather than as a public method win. The evaluated protocol reports the paper-facing model/backend, task count, metrics, and decoding budget for each condition.

A later selective ROI policy attempted to make the answer-blind crop tighter. It is not a scored public method because its candidate crops do not satisfy the pre-specified ROI inclusion criteria. The family summaries were OCRBench mean ROI area 0.756336 with 0.222222 full-frame fallback and 0.533333 infeasible rows, TextVQA mean ROI area 0.638715 with 0.333333 infeasible rows, and DocVQA mean ROI area 0.717082. The branch is therefore an exclusion boundary on selective ROI evidence, not a method column in Table 1.

The public rows answer a narrower sanity question than the controlled matrix. They check whether the evaluated VLM and OCR branches can run through real benchmark scorer paths on three public families, and whether prompt wording changes public aggregate scores. In this matrix, direct VLM is below prompt-only visual on OCRBench, TextVQA, and DocVQA, while prompt wording is most sensitive on TextVQA. PaddleOCR is also not monotonic across families, matching direct VLM on OCRBench but scoring much lower on the two VQA-style public slices. These facts support the paper’s scope boundary: the public matrix is useful evidence, but it is not a public intervention win.

The repair slices are reported with the same discipline. A hosted 120-task prompt/crop repair run and a Qwen2.5-VL-3B backend check are useful because they keep task ids paired while changing wording, crop context, or backend. They are not used to rewrite the main result. The main claim still comes from the completed four-condition controlled matrix, and the public claim comes from the three-family, five-condition public rows that were actually executed.

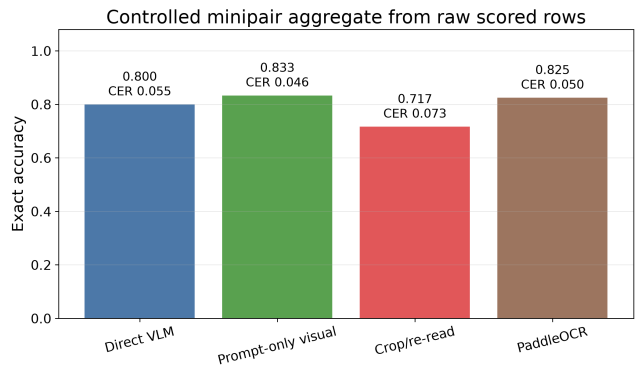


Figure 3: Condition-level exact accuracy and CER show the negative crop/re-read finding: 0.716667 exact versus 0.800000 direct VLM, 0.833333 prompt-only visual, and 0.825000 PaddleOCR.

Results

Table 1 gives the central controlled result and every confusion-family cell from the completed matrix. Table 2 gives the reviewer-facing setup/results view across the controlled diagnostic, three public families, answer-blind crop/re-read rows, and detector-only selective ROI exclusion boundary. Crop/re-read is the lowest controlled aggregate condition, while prompt-only visual is highest in this run; the row counts are balanced, with each controlled condition evaluated on the same 120 tasks.

Figure 3 visualizes the same aggregate pattern. The stable conclusion is that crop/re-read did not rescue exact glyph transcription under the completed matrix; the figure annotates the corresponding character-error rates. Crop/re-read is therefore reported as an evaluated negative baseline, not as an intervention win.

The paired task-level comparisons are negative for every comparator. Crop/re-read is lower than prompt-only visual by -0.116667 exact with bootstrap interval [-0.191667, -0.050000], lower than direct VLM by -0.083333 with interval [-0.141667, -0.033333], and lower than PaddleOCR by -0.108333 with interval [-0.175000, -0.050000]. The appendix restates these paired deltas before grouping the family-level losses.

The paired comparison sharpens the aggregate row. The crop branch is not behind because comparisons use different samples; each delta is paired on the same 120 task ids. Prompt-only visual is the strictest comparator because it changes only the instruction. Crop/re-read has 14 fewer exact matches than that condition, 10 fewer than direct VLM, and 13 fewer than PaddleOCR. These counts turn the decimal deltas into concrete errors. The crop intervention does not recover enough hard glyph cases to offset mistakes it introduces or leaves unresolved.

The intervals in the paired block are descriptive summaries over a fixed diagnostic matrix. They are not population claims about all fonts, VLM backends, or crop policies. They do guard against a weaker reading: crop/re-read is not approximately indistinguishable from the comparators

Table 1: Controlled exact-accuracy matrix: crop/re-read is below prompt-only visual by -0.116667, direct VLM by -0.083333, and PaddleOCR by -0.108333 on the same 120 generated tasks.

Family	Tasks	Direct VLM	Prompt-only	Crop/re-read	PaddleOCR
All	120	0.800	0.833	0.717	0.825
0/O	12	0.333	0.500	0.000	0.250
1/I	12	1.000	1.000	1.000	1.000
1/I	12	0.667	0.833	0.417	0.833
2/Z	12	0.667	0.750	0.667	0.667
5/S	12	1.000	1.000	1.000	1.000
8/B	12	0.833	0.750	0.833	0.917
O/O	12	1.000	0.917	1.000	1.000
c/e	12	1.000	1.000	1.000	1.000
Length edit	12	0.500	0.583	0.250	0.583
rn/m	12	1.000	1.000	1.000	1.000

Table 2: Cross-benchmark setup/results matrix: the controlled crop/re-read scaffold is negative, public answer-blind crop/re-read is mixed/no-win, and selective ROI has zero scored method rows.

Source	N / split	Backend	Budget	Metric	Result and boundary
Controlled minipair	120 ids	gpt-5-mini; PaddleOCR	32 tok	Exact	Prompt 0.833 , OCR 0.825, direct 0.800, crop 0.717; crop is lower than all comparators.
OCRBench public	45 ids	gpt-5-mini; PaddleOCR	64 tok	Hit	Prompt/plain 1.000 ; answer-blind crop/strict 0.978; direct/OCR 0.956.
TextVQA public	45 ids	gpt-5-mini; PaddleOCR	64 tok	VQA	Answer-blind crop 0.882 ; prompt 0.860; direct 0.838; strict 0.747; plain 0.696; OCR 0.022.
DocVQA public	45 ids	gpt-5-mini; PaddleOCR	64 tok	ANLS	Prompt 0.950 ; answer-blind crop 0.943; direct 0.941; strict 0.937; plain 0.915; OCR 0.000.
Selective ROI screen	135 rows	Detector gate	No VLM	Gate	Zero scored method rows; the detector-only policy fails the inclusion gate.

on these rows. All three intervals are entirely negative, and the largest upper endpoint is -0.033333 against direct VLM. Under the same-ID evidence standard, the measured scaffold remains a failed branch.

Table 2 reports the public matrix with the source, backend, budget, metric, and key score pattern visible in one place; the appendix gives the full visual summary. Each OCRBench, TextVQA, and DocVQA slice has 45 same-ID task ids with direct VLM, prompt-only visual, PaddleOCR, plain wording, strict wording, and a separate answer-blind crop/re-read evaluation. Direct VLM is below prompt-only visual on all three families. Answer-blind public crop/re-read is mixed versus prompt-only visual, and its confidence intervals include or touch zero, so Figure 5 is a no-win boundary rather than a positive public intervention result.

The attempted tighter public ROI route is kept outside the scored table. Its candidate crops do not meet the pre-specified selectivity and coverage criteria needed for inclusion as a scored public method. This keeps the selective public ROI branch separate from the answered benchmark rows and prevents it from supporting a selective-ROI or public method-win claim.

The two public prompt-wording variants are retained as numeric robustness checks rather than another body figure. TextVQA is the most sensitive slice: plain wording is -0.164444 and strict wording is -0.113333 relative to prompt-

only visual. OCRBench shifts are 0.000000 and -0.022222, and DocVQA shifts are -0.034271 and -0.012908. The result is a robustness finding, not a new winning method, because prompt-only visual remains strongest among the VLM prompt variants across all three public families. The appendix includes the corresponding prompt-sensitivity plot for readers who want the visual summary.

Analysis and Failure Cases

The failure taxonomy identifies where crop/re-read is most brittle. Figure 4 highlights the 0/O, 1/I, and length-edit slices. In the 0/O family, crop/re-read scores 0.000000 exact while direct VLM scores 0.333333, prompt-only visual scores 0.500000, and PaddleOCR scores 0.250000. In the 1/I family, crop/re-read reaches 0.416667 while prompt-only visual and PaddleOCR each reach 0.833333. In length-edit cases, crop/re-read scores 0.250000 while prompt-only visual and PaddleOCR score 0.583333.

The appendix tabulates the same family deltas. They show that the crop/re-read loss is concentrated rather than uniform: 1/I, 5/S, O/O, c/e, and rn/m tie the best non-crop score, while 0/O, 1/I, and length edits carry the aggregate loss. The hard rows are exactly where the diagnostic places character identity in conflict with lexical plausibility. In 0/O, crop/re-read gets no exact matches on the 12 tasks; in 1/I it gets 5 of 12 while prompt-only visual and PaddleOCR each

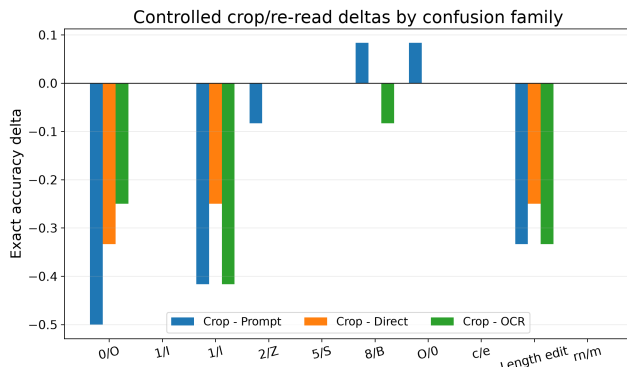


Figure 4: Failure-slice analysis localizes the crop/re-read loss to 0/O, 1/I, and length-edit examples rather than supporting a general verification gain.

get 10 of 12; in length edits it gets 3 of 12 while prompt-only visual and PaddleOCR get 7 of 12.

These slice counts identify the mechanism of the aggregate loss. Cropping does not uniformly degrade the image: it ties the best non-crop score on several families and therefore cannot be dismissed as a broken preprocessing step. The loss appears where the rendered target competes with a plausible lexical normalization. In 0/O rows, the visible digit and letter alternatives are close enough that the crop-only view does not preserve the intended identity. In 1/I and length-edit rows, the crop removes surrounding context but does not add a new constraint that forces exact character copying. The resulting error pattern is consistent with the paper’s diagnostic question: making the visual region closer is not the same as making the model faithful to the glyph sequence.

The paired aggregate tests confirm that the family pattern survives when each task id is counted once. Crop/re-read is lower than prompt-only visual by -0.116667 exact accuracy with a 95% bootstrap interval from -0.191667 to -0.050000 and McNemar $p = 0.002577$. It is lower than direct VLM by -0.083333 with interval $[-0.141667, -0.033333]$ and $p = 0.006348$, and lower than PaddleOCR by -0.108333 with interval $[-0.175000, -0.050000]$ and $p = 0.002350$. These are descriptive tests over the fixed diagnostic matrix, not estimates of stochastic training variation.

The error-type summaries support the same interpretation. In the 0/O family, crop/re-read makes digit-to-letter errors on all 12 rows, compared with 8 such errors for direct VLM, 6 for the prompt-only wording, and 9 for PaddleOCR. In the 1/I family, crop/re-read has 7 digit-to-letter errors, while prompt-only visual and PaddleOCR each have 2. In length edits, crop/re-read has the highest mean character error rate among the four conditions. These are not semantic disagreements about a natural-language answer; they are failures to preserve the generated target string.

The easy families are useful controls. Crop/re-read ties the best non-crop score on 1/I, 5/S, 0/0, c/e, and rn/m, which argues against a trivial explanation in which the crop image is always unusable or the scorer cannot credit crop/re-read. The hard-family losses are therefore more informative: the

method fails where visual character identity is most likely to compete with lexical plausibility.

The public matrix remains separate from the main controlled result. OCRBench is nearly saturated in this slice, while TextVQA and DocVQA expose a different family-dependent pattern: direct VLM is below prompt-only visual on all three families, prompt-wording shifts are largest on TextVQA, and PaddleOCR is much lower on the two VQA-style slices. The answer-blind public crop/re-read row adds one TextVQA slice-local positive delta but two negative family deltas, so the public evidence rules out public method-win language for the evaluated conditions. The subsequent detector-only selective ROI screen also rules out treating the tighter crop route as a hidden win: its candidate crops do not meet the ROI area, fallback, infeasibility, or detected-text coverage criteria needed for a scored public comparison. The bounded repair and Qwen2.5-VL checks are also treated as feasibility checks, not final evidence.

This separation matters because the public rows answer a different question from the controlled minipair rows. OCRBench, TextVQA, and DocVQA confirm that the same evaluation stack can produce real public scorer outputs across text recognition, scene-text VQA, and document VQA slices. They do not isolate a single rendered token or hold every visual nuisance variable fixed. The public pattern therefore constrains rhetoric rather than replacing the controlled finding: it prevents a direct-VLM public win claim, shows that prompt wording can shift scores without changing the image, and shows that answer-blind public crop/re-read is mixed/no-win overall. The controlled matrix remains the only evidence source in the paper where the crop branch, direct branch, prompt-only branch, and OCR baseline are all paired on identical hidden-gold strings.

The same-ID failure overlap is mainly an audit aid. OCRBench has 42 of 45 task ids solved by all five executed public conditions, while TextVQA and DocVQA include many PaddleOCR failures and prompt-variant failures. That pattern says the public rows are not simply copies of the controlled minipair distribution. It also prevents the paper from turning public evidence into a crop claim: the public crop/re-read rows are mixed, and prompt-only visual remains the strongest or tied strongest non-crop VLM prompt variant in aggregate on two of the three families.

The public matrix mainly guards against overfitting the narrative to a single synthetic artifact. It shows that the executed VLM route and PaddleOCR route produce scored rows under public OCRBench, TextVQA, and DocVQA protocols, and that direct VLM is below prompt-only visual on all three families. That negative public comparison is a constraint on the paper’s rhetoric: it prevents a public direct-VLM win claim and keeps the controlled minipair matrix as the central evidence. It also prevents a public crop/re-read win claim because the answer-blind run improves only TextVQA while losing on OCRBench and DocVQA.

The practical observation is therefore narrow. Visual focus is not automatically verification in this matrix: a crop can remove distracting pixels, but it can also remove context or scale cues that help a model preserve exact characters. A stronger crop policy remains plausible, but it would need to

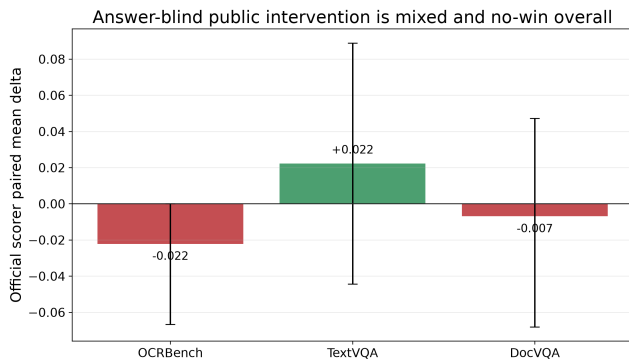


Figure 5: Answer-blind public crop/re-read deltas versus prompt-only visual are mixed/no-win overall: OCRBench is -0.022222, TextVQA is +0.022222, and DocVQA is -0.006980, with intervals including or touching zero.

improve the 0/O, 1/I, and length-edit rows while preserving the easy-family ties and while keeping gold strings out of prompts and filenames.

A broader paper would have a different evidence target. It would run larger official public splits, develop detector-only crop or zoom routes that first pass the failed selectivity gates, evaluate multiple VLM backends, and report prompt-family robustness. This note instead fixes a smaller question: under one completed controlled diagnostic, does a simple crop/re-read scaffold improve exact glyph transcription relative to the direct VLM, one prompt-only wording, and PaddleOCR? The measured answer is no.

Diagnostic Scoring and Evidence Boundary

The controlled minipair matrix uses deterministic gold transcription, so exact programmatic scoring is appropriate: a prediction either preserves the visible character sequence or it does not. This is stricter than answer-level VQA scoring, where a normalized token might still answer a downstream question. The public matrix uses official scorer paths for OCRBench, TextVQA, and DocVQA, but the 45-ID-per-family coverage is not enough for leaderboard language, public intervention-win claims, or full benchmark-scale validation.

The repair and backend slices obey the same boundary. They keep task IDs paired while varying prompt wording, crop context, or backend, but they do not replace the primary controlled matrix. A public benchmark paper would need larger official splits and benchmark-specific scoring across task families; this diagnostic note instead asks a narrower question about exact glyph preservation under controlled near-neighbor strings.

Figure 5 is the public counterpart to the controlled negative result. It shows that the answer-blind crop/re-read branch does not produce a stable public gain on the same three-family slice. The TextVQA improvement is paired with OCRBench and DocVQA losses, and the intervals include or touch zero. This pattern is why the paper treats public crop/re-read as boundary evidence instead of converting

the controlled diagnostic into a public intervention claim.

The selective ROI branch is an even stricter boundary. Its detector-only candidate crops do not meet the pre-specified area, fallback, infeasibility, or coverage criteria required for a scored public comparison. The paper therefore treats the selective ROI variant as evidence against a current selective-ROI win claim, not as an additional public score.

Conclusion

This paper reports a controlled diagnostic negative result. In a 480-row minimal-pair matrix, crop/re-read underperformed direct VLM transcription, one prompt-only visual wording, and PaddleOCR on exact in-image text transcription. The public matrix is bounded rather than decisive: direct VLM is below prompt-only visual on OCRBench, TextVQA, and DocVQA, prompt-wording sensitivity is largest on TextVQA, and PaddleOCR matches direct VLM only on OCRBench while falling much lower on TextVQA and DocVQA. The answer-blind public crop/re-read run is also mixed: it improves TextVQA by +0.022222 but is lower on OCRBench and DocVQA, with intervals including or touching zero. The finding narrows rather than broadens the claim: exact glyph reading remains fragile in this diagnostic, and a plausible train-free crop scaffold is not automatically a faithfulness improvement. A later detector-only selective ROI screen did not qualify as a scored public comparison, so broader public-harness, backend-general, selective ROI, or positive intervention claims remain outside the completed evidence.

Limitations and ethics. The study is limited to generated English-like strings and one hosted VLM backend. It does not evaluate handwriting, multilingual scripts, dense documents, natural-scene clutter, downstream question answering, or safety-critical OCR. The prompt-only visual condition is one baseline wording, the plain and strict public prompt variants are two measured robustness checks, crop/re-read is one crop policy, and public evidence is a 45-ID-per-family matrix whose answer-blind crop/re-read result is mixed/no-win overall. The benchmark uses generated rendered strings rather than personal documents, private images, or human-subject data. The main ethical risk is overstatement, so the paper reports the negative comparison and names the missing evidence. The study also does not establish deployment behavior under varied fonts, camera viewpoints, compression, document layout, or user-specific accessibility needs. The public rows are useful because they use real benchmark scorer paths, but 45 task ids per family cannot substitute for official full-split reporting. Future work should add larger public slices, detector-only ROI routes that first pass the failed selectivity gates, multiple crop and zoom policies, broader prompt-family robustness, and independent VLM backends before making stronger claims about verification or general visual text faithfulness. Until then, the responsible use of the result is diagnostic: it identifies one measured failure mode and one failed simple repair, not a guarantee about production OCR systems or a recommendation for safety-critical use. The same caution applies to accessibility and archival settings, where an OCR error can have practical consequences even if a benchmark

answer looks close. The artifact-level auditability in this paper is intended to make the negative result inspectable, but it should not be treated as a deployment validation protocol. The generated benchmark also should not be used as a substitute for user-facing readability tests. The strings are rendered to make hidden gold transcription unambiguous, not to represent the diversity of fonts, scripts, image capture conditions, or document layouts encountered by users. Any deployment-oriented study would need human readability checks, public split coverage, and application-specific error costs in addition to the exact-match scorer used here. Finally, the negative result should not be used to dismiss assistive visual text tools or specialized OCR systems. It only shows that one crop-only scaffold did not improve this paired diagnostic. Systems intended for users should be evaluated on the documents, languages, fonts, and error tolerances relevant to those users, with uncertainty surfaced rather than hidden behind an aggregate score. The paper also leaves several measurement questions open. It does not compare multiple hosted VLMs under the same public matrix, does not test whether a better detector-only public crop policy can produce an overall win, and does not measure whether padding, scale selection, or multi-crop voting changes the controlled hard-family rows. It also does not study calibration: the reported predictions are scored as strings, but the manuscript does not ask whether a model knows when it is uncertain about a glyph. These omissions matter because a practical verification system may need to abstain, ask for another view, or call a specialist OCR model rather than return a single normalized token. The present artifact supports row-level inspection of these failures, but it is not a complete design for such a system. The scorer itself is intentionally austere. Exact match is appropriate for the controlled strings because the gold target is known, but it cannot describe all downstream harms or tolerances. A future user-facing evaluation should separate minor typography differences, harmful substitutions, field-level document errors, and answer-level VQA failures. Keeping those categories separate would make the evidence more actionable than a single aggregate number, and it would avoid implying that every visual-text failure has the same consequence. This is especially important because visually small OCR mistakes can change names, amounts, medical abbreviations, or legal identifiers. The controlled examples in this paper are not drawn from those domains, and the study does not measure human recovery from model errors. A deployed assistant would need domain-specific review procedures, uncertainty reporting, and user override paths. The present diagnostic can help designers notice a failure mode, but it does not certify any model or workflow for consequential document reading. For that reason, the paper’s artifact bundle is best read as a reproducible failure analysis package. It supports checking which rows failed, which families carried the loss, and which claims were deliberately not made. It does not replace external validation by communities who depend on visual-text systems.

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Reproducibility Checklist

Task and data. The controlled diagnostic contains 120 generated minimal-pair rendered-string tasks across ten confusion families. The public matrix contains 45 public task ids per family for OCRBench, TextVQA, and DocVQA, with 675 scored public rows across direct VLM, prompt-only visual, PaddleOCR, plain prompt wording, and strict prompt wording. The answer-blind public crop/re-read evaluation adds 405 scored rows across the same three families and the direct, prompt-only, and answer-blind crop/re-read conditions. The later selective ROI screen is reported only as an exclusion boundary because its candidate crops do not satisfy the inclusion criteria for scored public comparison. Larger official public slices and positive public crop/re-read evidence are not available in this study.

Task construction. Each generated task contains one rendered string and one hidden gold transcription. The visible string is the only target: there is no surrounding question, document layout, or semantic context that could justify a paraphrase. This is the reason the benchmark can use exact character identity as the primary outcome. The ten confusion families are balanced at 12 tasks each, which makes family-level comparisons interpretable without reweighting.

Evaluated systems. Hosted VLM conditions use the paper-facing ‘gpt-5-mini’ model identifier with temperature

zero and 32 output tokens. PaddleOCR 3.x is used as the OCR baseline.

Condition replay. The direct VLM and prompt-only visual conditions use the original rendered image. The prompt-only branch changes the instruction but does not crop the image. The crop/re-read branch changes the image payload to the deterministic glyph-region crop while keeping the hosted VLM decoding budget fixed. PaddleOCR uses the same rendered task images and supplies the OCR-specialist comparison. Because every condition is evaluated on the same 120 task ids, the paired deltas compare rows rather than independent samples.

Metrics. Exact transcription accuracy and character error rate are computed per row. Paired comparisons use task-level exact-match differences, bootstrap intervals, and McNemar exact tests.

Statistical interpretation. The bootstrap intervals and McNemar tests are descriptive summaries over this fixed diagnostic matrix. They show that the negative crop/re-read deltas are not produced by an arbitrary aggregate presentation, but they do not estimate variation over fonts, public benchmark distributions, model families, or prompt families. Those sources of variation require the follow-up ablations named above.

Reproducibility materials. The reproducibility bundle contains task images, prompts, predictions, scored rows, tabular summaries, and data figures. No training or parameter update is used.

Artifact availability. The controlled diagnostic is designed to be inspectable at row level. The artifact bundle contains the rendered task images, task metadata, prompts, predictions, scored rows, aggregate tables, paired-comparison summaries, failure-taxonomy summaries, and the plotted data figures used in the paper. Those materials are sufficient to reproduce the reported tables without new model calls: the scorer consumes gold strings and predictions, groups rows by condition and confusion family, and recomputes exact accuracy and character error rate. The OCRBench/TextVQA/DocVQA public rows are included as separate public-scorer evidence and are not merged into the controlled matrix. Additional public-harness evidence would need separate benchmark-specific artifacts with official split identifiers, scoring protocol, and result tables rather than being merged into the controlled matrix.

Paper-facing replay boundary. The manuscript reports the model or OCR backend, condition role, task count, decoding budget, metric, and numerical takeaway for each completed condition. It intentionally does not turn local execution details into scientific claims. The replay materials are meant to let a reviewer inspect the task images, prompts, predictions, and scorer outputs that produce the tables and figures.

Scorer auditability. Each row has enough information to recompute the reported metrics without requerying a model: task id, condition, prediction, gold transcription, exact-match score, character error rate, and confusion family. The aggregate table is a direct grouping of those rows by condition and family. The paired comparisons align rows by task id before computing differences, so a reviewer can

inspect which examples change status between crop/re-read and each comparator. This row-level structure is important for a negative result because it rules out a presentation that compares different task subsets across conditions.

Gold visibility. Gold strings are used by the scorer, not by the evaluated prompts. The paper-facing task shown to each evaluated condition is single-string transcription from an image. This distinction matters because the benchmark deliberately contains near-neighbor strings that invite auto-correction. If a model predicts a common-looking token instead of the rendered target, the scorer marks the row wrong even when the prediction is linguistically plausible.

Negative-result handling. The original crop/re-read hypothesis was that a tighter visual region might help exact transcription. The completed matrix did not support that hypothesis. The reproducible artifact therefore keeps crop/re-read as an evaluated baseline rather than relabeling the prompt condition as a replacement method. This convention keeps the reported claim aligned with the measured comparison and keeps the remaining evidence gaps explicit.

Known gaps. The study lacks broader prompt-family robustness, larger public benchmark slices, a winning public crop/re-read result, and a crop-quality/context study that changes the negative controlled finding.

Appendix: Additional Diagnostic Details

Public Matrix

The public matrix is included to bound the controlled diagnostic, not to turn it into a public method-win claim. The same 45 task ids per family are evaluated under direct VLM, prompt-only visual, PaddleOCR, and two prompt-wording variants. Figure 6 shows that prompt-only visual remains above direct VLM on all three public families, while PaddleOCR depends strongly on the benchmark family.

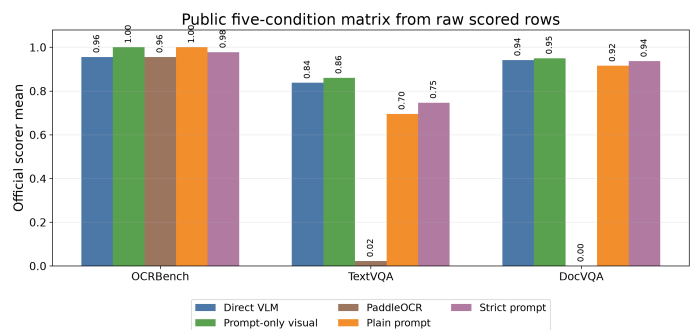


Figure 6: Public five-condition matrix: direct VLM is below prompt-only visual on OCRBench, TextVQA, and DocVQA, while PaddleOCR is family-dependent.

Prompt Wording Sensitivity

The prompt variants use the same images as the public prompt-only visual condition, so their shifts isolate wording sensitivity rather than a new visual intervention. Figure 7 shows the largest movement on TextVQA: plain wording shifts by -0.164444 and strict wording by -0.113333 relative to prompt-only visual.

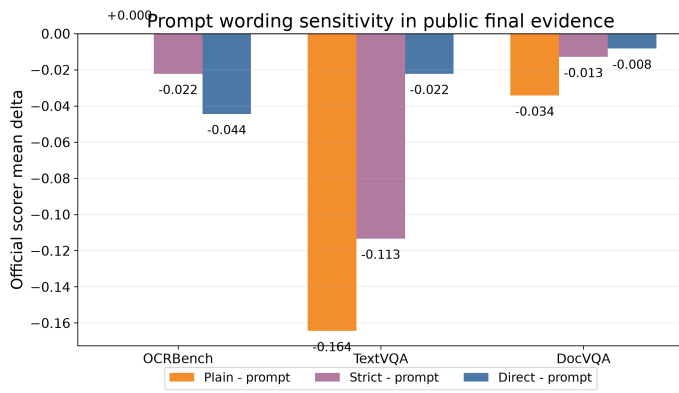


Figure 7: Public prompt wording sensitivity: TextVQA moves most, while prompt-only visual remains the strongest VLM wording in this public matrix.

Controlled Failure Slices

The controlled family breakdown explains where the aggregate crop/re-read loss comes from. The largest gaps are concentrated in 0/O, 1/l, and length edits; five easier families tie the best non-crop comparator. The paired task-level deltas are negative against every comparator: -0.116667 versus prompt-only visual with interval [-0.191667, -0.050000], -0.083333 versus direct VLM with interval [-0.141667, -0.033333], and -0.108333 versus PaddleOCR with interval [-0.175000, -0.050000].

Table 3: Controlled failure-slice summary: the largest crop/re-read losses are 0.500 on 0/O, 0.417 on 1/l, and 0.333 on length edits, while five easier families tie the best comparator.

Family	Crop exact	Best non-crop	Gap
<i>Hard-family losses</i>			
0/O	0.000	0.500	-0.500
1/l	0.417	0.833	-0.417
Length edit	0.250	0.583	-0.333
<i>Smaller losses and ties</i>			
1/I	1.000	1.000	0.000
2/Z	0.667	0.750	-0.083
5/S	1.000	1.000	0.000
8/B	0.833	0.917	-0.083
O/O	1.000	1.000	0.000
c/e	1.000	1.000	0.000
rn/m	1.000	1.000	0.000

The grouped table shows why the aggregate loss is concentrated rather than uniform. Crop/re-read ties the best non-crop comparator on five easier families, and its smaller losses on 2/Z and 8/B do not offset the three hard-family failures. For transparency, the paper keeps the failed crop/re-read scaffold alongside the stronger comparators so the negative result can be inspected at aggregate, paired-delta, family-loss, and error-type levels.